

NANYANG JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

Mark Scheme

CLASS

BIOLOGY

9648/02

Paper 2 Core Paper

September 2015

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
8	
Section B	
Total	100

This document consists of **23 printed pages and no blank page.**

[Turn over

Section A

Answer **all** the questions in this section.

- 1 **Fig. 1.1** is a computer-generated image of the enzyme hexokinase binding with its substrate, glucose. The product of the enzyme-catalysed reaction is glucose-6-phosphate.

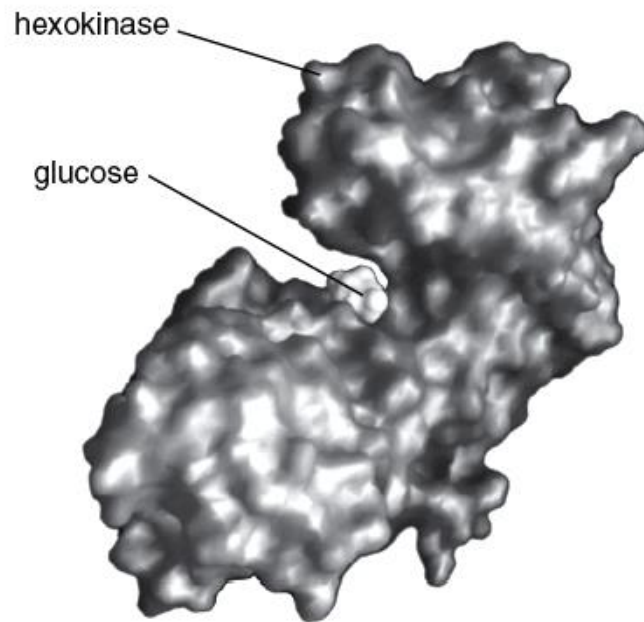


Fig. 1.1

- (a) Hexokinase binds with glucose using the induced fit mechanism.

Describe how an enzyme-substrate complex forms by this mechanism.

glucose/ substrate, is not complementary / is partially complementary, to active site ;

Enzyme's active site changes shape and fits around, when substrate, enters / binds ;
 ® if substrate/ glucose changes

stronger binding of substrate to active site ;

further detail ; e.g. becomes complementary to/ fits more tightly to, glucose/ substrate
 interaction of, functional groups /R-groups / side-chains formation of (named) bond
 but not disulfide or peptide bond

[3]

- (b) Suggest how enzymes which use the induced fit mechanism can be less affected by competitive inhibitors than those which use the lock and key mechanism.

(competitive) inhibitor has, same/ similar, shape to substrate therefore binds to active site ;

inhibitor does not induce the same change in, 3D shape/ tertiary structure/ active site
 (as the substrate) / so (inhibitor) less likely to bind (successfully) in active site doesn't
 fit well;

in lock and key, the inhibitor, fits directly into/ is perfect complementary to/ binds to,
 active site ;

idea that because it does not have same functional groups (in same positions)/AW ;

[2]

- (c) Glucose can enter cells by active transport or facilitated diffusion.
Glucose-6-phosphate is a molecule that cannot move out of cells.

(i) Describe two **differences** between active transport and facilitated diffusion.

active transport requires, ATP/ energy (whereas facilitated diffusion does not) ;

active transport moves substances against the concentration gradient whereas facilitated diffusion moves substances down the concentration gradient ;

active transport uses only carrier proteins (whereas facilitated diffusion uses both carrier and channel proteins) ;

@ active transport can involve cotransport but facilitated diffusion does not

[2]

(ii) Suggest why glucose-6-phosphate cannot move out of cells.

too large/ too big to pass through temporary gaps in the phospholipid bilayer ; R 'it is a big molecule' unqualified

Polar / charged, so cannot pass through hydrophobic region of membrane / fatty acid tails ;

no, specific /AW, protein, in membrane/ carrier/ channel / no protein for G-6-P

AVP ; e.g. gated channels are closed

[2]

[Total: 9]

- 2 **Fig. 2.1** is a section through the rough endoplasmic reticulum representing the synthesis of proteins.

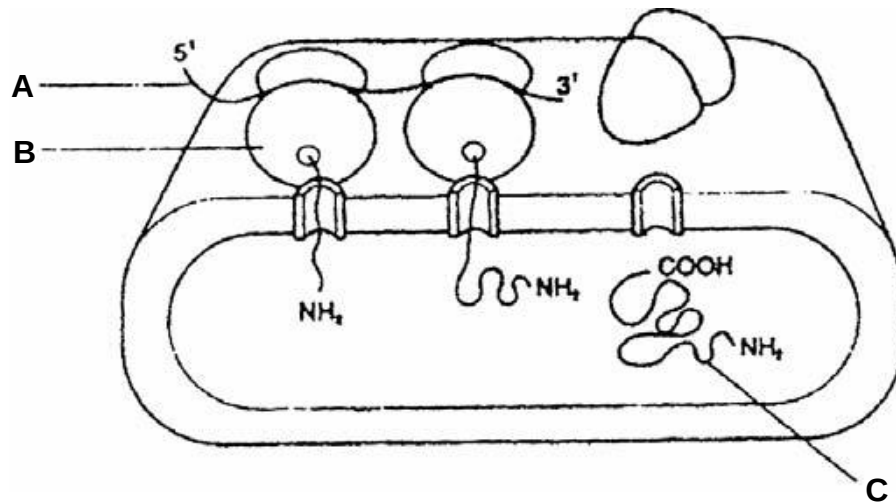


Fig. 2.1

- (a) Name the structures A to C.

A messenger ribonucleic acid;

B ribosome/ large subunit of ribosome ;

C polypeptide / protein ;

[3]

- (b) Describe and explain the role of structure B in protein synthesis.

Ribosome uses mRNA as a template for translation forming polypeptide / protein;

rRNA of the large subunit has the peptidyl transferase activity that catalyses peptide bond formation;

Consists of P and A site which allows amino-acyl tRNA to fit into the ribosome;

rRNA is involved in binding to mRNA and tRNA;

[3]

Fig. 2.2 is a diagram of a section of mRNA showing the sequence of three of the codons.

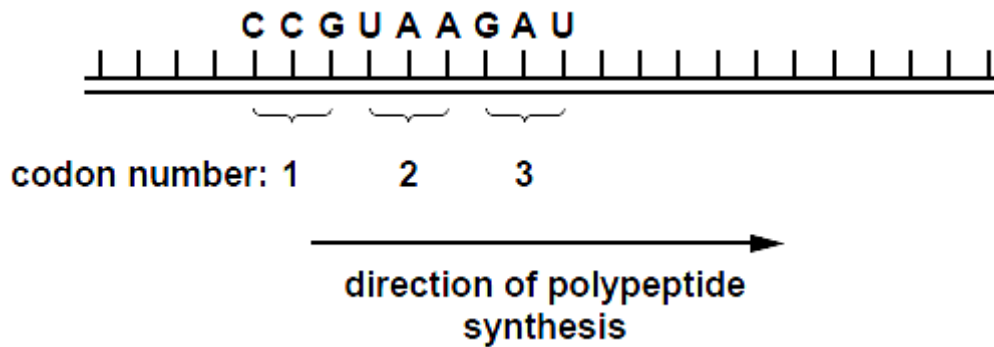


Fig. 2.2

(c) State the base sequences of

(i) the tRNA anticodon complementary to codon 1.

GGC;

[1]

(ii) the DNA sequence which coded for codon 3.

CTA;

[1]

(d) Explain the relationship between the sequence of the DNA triplets and the structure of the hormone.

Sequence of nucleotides / bases on DNA determines sequence of amino acids in polypeptide (i.e. primary structure);

Different amino acids have different R groups, therefore forming different bonds, resulting in unique three-dimensional conformation / structure;

[2]

[Total: 10]

- 3 Viruses such as bacteriophages have been described as being “organisms at the edge of life”.

(a) State one characteristic of viruses that may classify them as being

(i) living.

Can reproduce when it takes over the genetic machinery of its host cells to replicate more viral particles;

Adaptable as viruses have high mutation rates so as to have genetic variation and have a higher chance of survival in unfavourable environments;

Able to carry out metabolic activity as viruses direct its host cells to provide the energy needed to produce more viral particles;

[1]

(ii) non-living.

They do not have cellular organization;

They are unable to acquire and use energy in order to maintain metabolic processes for survival on their own;

They do not grow and develop;

They are unable to reproduce on their own;

And therefore are unable to adapt to the environment without help of a host cell;

They do not have specialized receptors that detect environmental stimuli to allow their cells to adjust metabolism in response;

[1]

Lambda (λ) phage infections result in the lysis of the bacterial host cells which shows up as a plaque or a zone of clearing on an agar plate containing a dense growth of bacteria. This observation can be used to quantify the number of infectious bacteriophage particles through a plaque assay. 'Plaque Forming Unit (PFU)' gives the measurement of the number of infectious viral particles present during the experiment.

Fig. 3.1 below shows the changes in PFU caused by lambda (λ) phages infecting E.coli cells upon irradiation by UV.

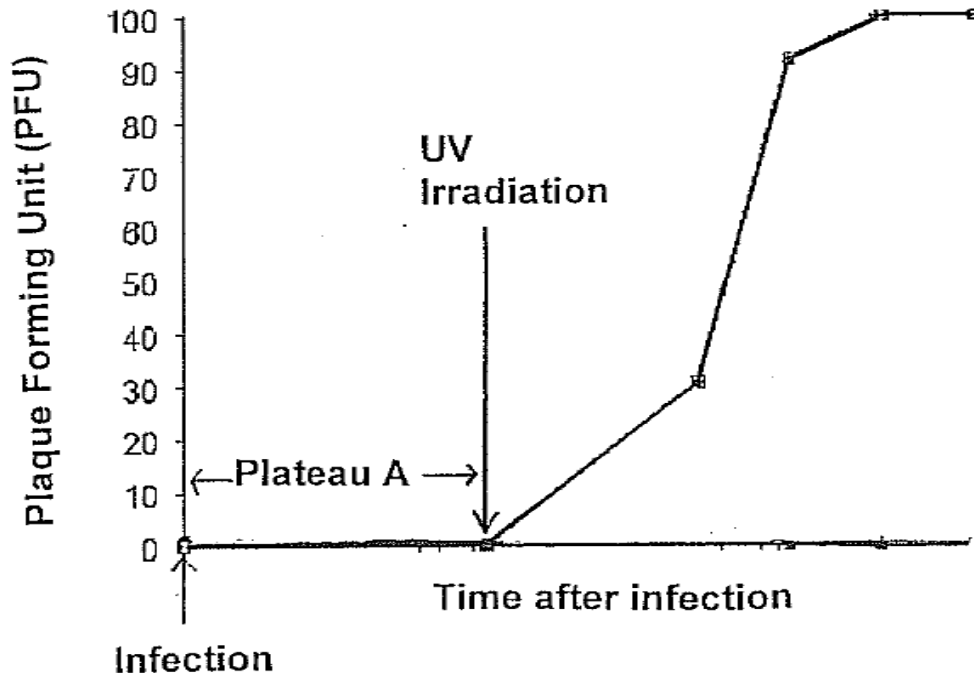


Fig. 3.1

(b) Briefly describe the structure of a lambda (λ) phage.

The head is an icosahedral protein capsid;

Enclosing a linear double-stranded DNA as its genome;

Attached to the head is the tail sheath with fibres; @ tail shaft;

[3]

(c) Suggest an explanation for the flat part of the graph, labelled A, as shown in **Fig. 3.1**.

Phage lambda is undergoing lysogenic cycle with its phage DNA integrated in bacterial host DNA as prophage;

A prophage gene code for a protein / repressor that prevents transcription of most of the other prophage genes, allowing phage genome to remain dormant within the bacterium;

No viral DNA and proteins synthesised for assembly to form new phages and no lysis occurs so zero PFU;

[2]

(d) Account for the rise in PFU shortly after UV irradiation.

UV irradiation induce the phage to transit from lysogenic cycle to lytic cycle / inactivates repressor protein, resulting in expression of prophage genes;

Prophage is excised from the bacterial host chromosome and phage enters lytic cycle;

Phage DNA directs synthesis of phage proteins and replication of phage DNA by the host cell machinery;

Assembly of viral components into new phages and subsequent release of phages by lysis, resulting in rise in PFU from 0 to 100;

[3]

[Total: 10]

- 4 The Ras protein, encoded by the Ras proto-oncogene, is a G protein that relays a signal from a growth factor receptor on the plasma membrane to a cascade of protein kinases. The conversion of the Ras proto-oncogene to Ras oncogene causes the latter to act in a dominant manner. The permanently activated protein directs the cell to grow and divide in the absence of external signals, leading to uncontrolled cell division and the formation of a tumour.

Somatic mutations in the H-ras gene in bladder cells have been associated with bladder cancer. One specific gene mutation has been identified in a significant percentage of bladder tumors. A segment of the gene is shown in **Table 4**.

Table 4

codon number	9	10	11	12	13	14	15
normal Human H-Ras	GUG Val	GGC Gly	GCC Ala	GGC Gly	GGU Gly	GUG Val	GGC Gly
oncogenic H-Ras	GUG Val	GGC Gly	GCC Ala	GUC Val	GGU Gly	GUG Val	GGC Gly

- (a) With reference to **Table 4**, explain how this mutation would affect the **function** of the protein formed.

Mutation result in a change in amino acid from glycine to valine at codon 12;

Result in different R-group interactions, distorting the specific 3-dimensional conformation / tertiary structure;

Idea of permanent binding of GTP to G-protein / GTPase activity within G-protein not functional

[3]

- (b) Explain what is meant by “the Ras oncogene acts in a dominant manner”.

Gain of function mutation of ras oncogene;

Presence of a single copy of the oncogene / mutated proto-oncogene is sufficient to mask the effect of the normal allele / proto-oncogene;

[2]

- (c) Besides gene mutation, describe one way in which a proto-oncogene may be converted to oncogene.

Gene amplification: DNA polymerase slips from the DNA template strand and reattach at a site that was already replicated;

Increase the number of copies of proto-oncogene and excess protein are produced;

OR Translocation: chromosomal fragment breaks and joins with a non-homologous chromosome;

Proto-oncogene is under the control of an more active promoter / enhancer which results in overproduction of the normal protein;

[2]

- (d) The expression of H-Ras gene can be furthered controlled at post-translational level.
Explain how gene expression may be controlled at post-translational level.

The protein is activated by biochemical modification / attaching other biochemical functional groups; (such as GTP)

Structural modification such as removal amino acids from the protein may occur to produce the mature protein;

Ubiquitin are covalently attached to unwanted / excess / misfolded proteins which are degraded by proteasomes;

[3]

- (e) Many individuals are afflicted with cancers when they are past 40 years old. Suggest why these cancers do not normally appear in younger individuals.

Cancer is a multi-step process which takes;

Time for accumulation of mutations in tumour suppressor genes and proto-oncogenes;

Time for reactivation of telomerase gene, tumour angiogenesis, invasion and metastasis (any two to score 1m);

[2]

[Total: 12]

5

- (a) Explain what is meant by the term *heterozygous genotype*.

heterozygous:

Two different alleles for a gene at a particular locus on a pair of homologous chromosomes

/ different allele pair for a gene / AW;

Produces gametes with different genotypes ;

genotype:

Alleles present in an organism / particular alleles of a gene / genetic constitution / AW;

[2]

- (b) The budgerigar, *Melopsittacus undulatus*, is a small type of parrot that is native to Australia.

Fig. 5.1 shows a budgerigar.



Fig. 5.1

A budgerigar can have blue, green, yellow or white feathers.

Two genes, **A / a** and **D / d**, are involved in the inheritance of feather colour in budgerigars.

- A bird has at least one dominant allele **A** but is homozygous for **d** has blue feathers.
- A bird which has at least one dominant allele **D** but is homozygous for **a** has yellow feathers.
- A bird with at least one dominant **A** allele and one dominant **D** allele has green feathers.
- A bird that is homozygous for **a** and **d** has white feathers.

Two green-feathered budgerigars, heterozygous at both gene loci, were crossed.
Draw a genetic diagram of this cross to show the probability of producing offspring with yellow feathers.

Parental generation

Parental phenotypes Green feathers x Green feathers

Parental genotypes AaDd AaDd

After meiosis,
gametes produced (AD) (Ad) (aD) (ad) (AD) (Ad) (aD) (ad) ;

By random fertilization,

	(AD)	(Ad)	(aD)	(ad)
(AD)	AADD Green feathers	AADd Green feathers	AaDD Green feathers	AaDd Green feathers
(Ad)	AADd Green feathers	AAdd Blue feathers	AaDd Green feathers	Aadd Blue feathers
(aD)	AaDD Green feathers	AaDd Green feathers	aaDD Yellow feathers	aaDd Yellow feathers
(ad)	AaDd Green feathers	Aadd Blue feathers	aaDd Yellow feathers	aadd White feathers

Offspring generation

Offspring genotypic ratio 1AADD, 2AADd, 2AaDD, 4AaDd 1AAdd, 2Aadd 1aaDD, 2aaDd 1aadd

Offspring phenotypic ratio 9 Green feathers 3 Blue feathers 3 Yellow feathers 1 White feathers ;

Probability of producing offspring with yellow feathers: 3/16;

[4]

(c) State the name for this type of interaction between gene loci.

Epistasis;

[1]

- (d) Explain how you would determine that a budgerigar with green feathers is homozygous dominant for both gene loci.

Mate / cross the bird with green feathers with another bird with white feathers i.e. aadd;

.....
If the cross results in offspring with all green feathers, then the bird (parent) must be homozygous dominant for both gene loci;

.....
~~If offspring with blue, yellow or white feathers are observed, the bird (parent) must have one recessive allele at least one gene loci;~~

.....
[2]

[Total: 9]

6 Fig. 6.1 outlines some steps in glucose metabolism in mammalian cells.

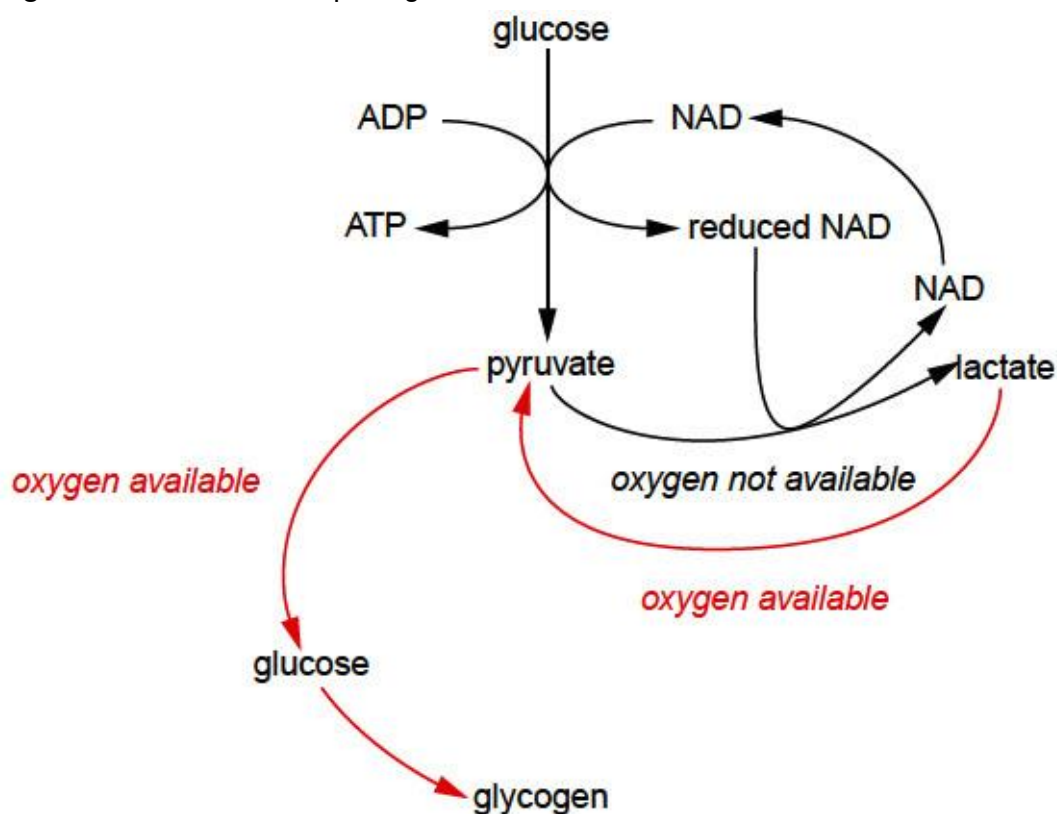


Fig. 6.1

(a) With reference to Fig. 6.1,

(i) explain why, in the absence of oxygen, pyruvate needs to be converted to lactate;

regenerate NAD^+ so that glycolysis can continue;

to produce (2) ATP via substrate level phosphorylation;

[2]

(ii) name the type of reaction **and** the type of bonds formed when glucose molecules are used to make glycogen.

Type of reaction: condensation

Type of bonds: glycosidic bonds (α -1,4- and α -1,6-)

[1]

- (b) ATP is often known as the universal 'energy currency'.
Outline how ATP is suitable for acting as an energy currency.

terminal phosphate removed easily ;
 releasing, useful / AW, amount of energy when hydrolysed ;

 allows efficient energy transfer ;
 small / water soluble, easily moved around cell ;

 is used by (many), enzymes / proteins ;
 idea that adenine + ribose = recognition site;

 idea that easily recycled ;
 three phosphate groups are negatively charged and unstable ;
 charged so does not pass through membranes (and lost to cell) ;
 AVP ;

[2]

- (c) Most ATP is made in cells by membrane systems that create proton gradients by pumping protons from one compartment to another.

Fig. 6.2 shows two such membrane systems.

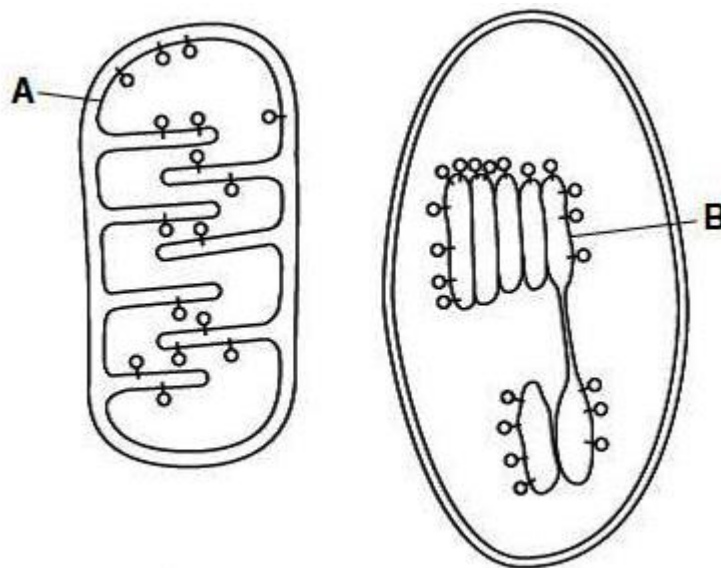


Fig. 6.2

- (i) Draw arrows onto each of the membrane systems in Fig. 6.2 to show the direction in which protons are pumped.

[1]

A – matrix to inter membrane space,
 B – into thylakoid,

® if goes through synthase

- (ii) Outline how energy is made available for pumping protons across such membranes.

Oxidation of glucose via glycolysis and Krebs cycle/ absorption of light energy by pigments results in formation of reduced coenzymes;

Electrons from reduced coenzymes/ NAD/ FAD/ NADP passed down electron carriers each with an energy level lower than the one preceding it;

Energy released by transfer of electrons used to pump H^+ from stroma/ matrix to thylakoid space/ intermembrane space;

energised electrons ;

(A) from oxidation of, food / named substrate(s) ;

from oxidation of reduced, hydrogen carriers / named ;

ref redox potential of reduced, NAD ; A ref FAD

(B) from absorption of light by pigments ;

electron transport chain ;

protein complexes with decreasing redox potential / AW ;

final acceptor for, electrons / hydrogen ;

[3]

- (iii) Explain how ATP synthase is involved in the production of ATP.

Diffusion of H^+ down the gradient through stalked particles containing ATP synthase;

Catalysing the formation of ATP from ADP + P_i

proton gradient / AW ;

facilitated diffusion ;

kinetic energy ;

phosphate added to ADP ;

detail of structure of ATP synthase ;

detail of function of ATP synthase ;

[2]

[Total: 11]

- 7 Serotonin is a neurotransmitter which is produced by certain neurons in the brain. One of its effects is to increase the activity of sensory neurons in the brain. It also usually improves a person's mood and keeps the person awake.

Fig. 7 below shows a synapse which serotonin is the neurotransmitter.

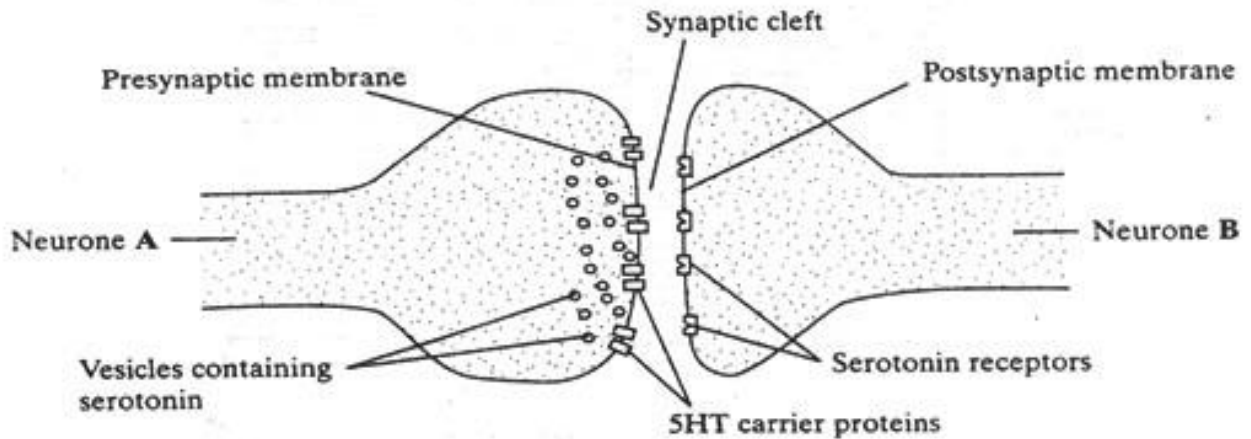


Fig. 7

(a)

- (i) Describe how the neurotransmitter is released into the synaptic cleft.

Arrival of impulse /action potential/ membrane depolarization → influx of Ca^{2+}

R: stimulus

Vesicles move toward the presynaptic membrane and fuse with the membrane → Neurotransmitters released via exocytosis

R: bind to membrane

[2]

- (ii) Explain how the release of the neurotransmitter, serotonin, results in the transmission of an impulse from neuron A to neuron B.

Serotonin diffuse across the synaptic cleft to the postsynaptic membrane;

Binds to complementary ligand-binding site on serotonin receptors/ligand-gated Na^+ channel

R: active site on receptor

This causes increase permeability of the postsynaptic membrane to Na^+ ions / opening of ligand-gated Na^+ channels and influx of Na^+ ions;

R: opening of serotonin receptor

Local depolarization of the postsynaptic membrane occurs; (inside becomes more +ve)

(FYI: The built up of local depolarization at the axon hillock due to summation that exceeds threshold potential results in generation of an action potential of neuron B)

[4]

Serotonin is normally rapidly reabsorbed from the synaptic cleft by 5HT carrier proteins in the presynaptic membrane.

(b) Suggest an advantage of rapidly absorbing the serotonin.

- This prevents overstimulation/continuous stimulation of the postsynaptic membrane by the stimulus from neuron
- Allows transmission of impulse to be rapidly terminated
- Allows further/faster transmission of another signal across the synapse (R: general statements about increasing rate of transmission without reference to a new signal)
- Allow serotonin to be recycled for rapid formation of new signal

[1]

The active ingredient in the drug, Ecstasy, is MDMA. It blocks the attachment of serotonin molecules to the 5HT carrier proteins.

(c) Suggest how MDMA may temporarily improve a person's mood

Serotonin stays in synaptic cleft/ attached to receptors and not absorbed.

This produces continual stimulation / depolarization of the postsynaptic membrane. R: Action potentials are generated more frequently.

Leads to impulses stimulating the sensory neurons in the brain which improves the person's mood for a prolonged period.

[2]

[Total: 9]

- 8 There are over 40 Galapagos Islands including the small and isolated island named Daphne Major. Only two species of Darwin finches are found on this island. From 1975 to 1977, Daphne Major experienced a severe drought. A ground finch, *Geospiza fortis*, feeds on seeds on Daphne Major.

Fig. 8.1 shows the ground finch, *G. fortis*.



Fig. 8.1

- One of the few plants that survived the drought produced large seeds inside tough fruits.
 - Many *G. fortis* died during the drought and the population declined from 1400 in 1975 to 190 in 1977.
 - The mean beak depth of those *G. fortis* that died was 10.68 mm and the mean beak depth of those that survived was 11.07 mm.
 - The mean beak depth of *G. fortis* before the drought was 10.86 mm.
- (a) Calculate the percentage decrease in population size between the years 1975 and 1977. Show all the steps in your calculation and give your answer to the nearest whole number.

$$(1400 - 190) / 1400 * 100 = 86\%$$

@ -86%

[2]

- (b) Describe the environmental factors that act as forces of natural selection on finches during drought years.

Drought resulted in limited food / reduced availability of food.

The type of food available changed from small seeds to large seeds.

Competition for food.

[2]

- (c) Suggest why some *G. fortis* were able to survive the drought while others died.

Drought reduced available food or fewer small seeds produced ;

Finches with larger beaks survived or finches with smaller beaks died ;

Finches are able to open tough fruits / able to feed on larger seeds ;

tough fruit / size of seed, acted as selection pressure ;

[3]

Natural selection was taking place on Daphne Major.

- (d) Suggest why the type of natural selection operating on *G. fortis* **during** the drought is directional selection.

Explanation: selection pressure acts on one extreme (of range) ;

[1]

- (e) Outline the mechanisms that may have let natural selection lead to the evolution of the thirteen species of Darwin finches now found on the Galapagos Islands.

ref. isolation of islands ;

ref. different environmental conditions / selection pressures on different islands ;

ref. adaptive radiation ;

ref. stabilizing selection ;

[2]

[Total: 10]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 9 (a)** Outline the arrangements and functions of named membranes within the cell. [7]
(b) Explain how the fluid mosaic model of cell surface membrane facilitates the transport of substances. [7]
(c) Describe the main stages of cell signaling using a named example. [6]

[Total:20]

- 10 (a)** Describe how the prokaryote, E.coli, is able to respond to varying concentrations of lactose and glucose [8]
(b) Explain how genetic variation in bacteria could have arisen. [8]
(c) Suggest why the influenza virus displays high genetic diversity. [4]

[Total:20]

9 (a) Outline the arrangements and functions of named membranes within the cell.

1. Nuclear envelope: double membrane structure with the presence of nuclear pores;
2. Encloses the DNA within the nucleus / nuclear pore allow for transport of mRNA from nucleus to cytoplasm;
3. Mitochondria: double membrane structure with the electron transport chain and stalked particles containing ATP synthase embedded in the cristae;
4. Allow for the synthesis of ATP when protons diffuse down the electrochemical proton gradient generated across the cristae.
5. Chloroplast: double membrane structure with the electron transport chain and stalked particles containing ATP synthase embedded in the thylakoids;
6. Allow for the synthesis of ATP through photophosphorylation for the reduction of glycerate phosphate to triose phosphate in Calvin cycle.
7. Endoplasmic reticulum: single membrane + transport vesicles bud off for intracellular transport / contain transport protein for polypeptide to enter lumen for folding into proteins;
8. Golgi apparatus: single membrane + transport vesicle from ER fuses with cis face;
9. And after chemical modification of proteins, secretory vesicles bud off trans face;
10. Vesicles contains proteins to be secreted out of cell by fusing with cell surface membrane through exocytosis;
11. Lysosomes: single membrane bound + contain hydrolytic enzymes;
12. Fuses with endosome for digestion of foreign particles;

9(b)

1. Cell surface membrane are made of phospholipids which are amphipathic;
2. Fluid refers to phospholipids and proteins can move freely;
3. Usually laterally within the same monolayer and seldom do a flip-flop;
4. Mosaic refers to proteins molecules are scattered and embedded within phospholipid bilayer;
5. Presence of transmembrane proteins spanning across both layers for transport of molecules;
6. Partially permeable membrane allows transport of certain molecules;
7. Hydrophobic tails repel polar substances like charged ions and close packing of hydrophobic tails prevents large molecules like glucose from passing through;
8. It allows the passage of hydrophobic molecule like oxygen and relatively small hydrophilic molecules such as water;
9. Membranes are impermeable to ions. Membrane allows transport of specific solutes through transport proteins. These proteins transport substances, like charged ions and large polar molecules like glucose, which do not readily pass through cell membrane;
10. Endocytosis is the process of intake of substances by formation of vesicles via invagination of cell surface membrane. A small area of plasma membrane sinks inwards and pinches off to form a vesicle containing the substance.
11. Exocytosis involves transport of materials out of cell. Materials to be transported out are enclosed in a secretory vesicle, usually from Golgi apparatus. Vesicles move along cytoskeleton to cell surface membrane and vesicle membrane fuses with it. Contents are then discharged from cell.

Insulin-RTK cell signalling

1. Ligand-receptor interaction: Binding of insulin to receptor tyrosine kinase (RTK) / tyrosine kinase receptor (TKR) forms a ligand-receptor complex, which will cause two RTK molecules to dimerise;
2. Contact between the two adjacent RTK tails activates their tyrosine kinase function;
3. This result in each RTK tail phosphorylating the tyrosine residues found on each other / autophosphorylation;
4. Activated RTK will trigger the assembly of adaptor proteins on the receptor tails, which will further recruit and activate other downstream relay molecules / protein kinase;
5. Signal Transduction: Each activated protein kinase will initiate a sequential phosphorylation and activation of other kinases, resulting in a phosphorylation cascade;
6. At each phosphorylation step, each activated kinase is able to activate a large number of the next relay molecule / kinase;
7. At each catalytic step in the cascade, number of activated product is always greater than those in the preceding step, resulting in signal amplification;
8. Large number of glycogen synthase is activated;
9. which will catalyse for the synthesis of glycogen from glucose / glycogenesis;
10. Cellular response: Therefore, binding of 1 insulin to receptors will lead to synthesis of large amount of glycogen /
Vesicles embedded with glucose transporters to migrate towards the cell surface membrane and facilitate the transport of glucose into the liver / muscle cells;

Glucagon-GPLR cell signalling

11. Ligand-receptor interaction: Binding of glucagon to G protein-linked receptor (GPLR) / G protein-coupled receptor (GPCR) forms a ligand-receptor complex, which will cause GPCR to undergo a conformational change;
12. This activates G protein that is closely associated with the GPCR and causes the G protein to exchange GDP for GTP;
13. G-protein will translocate and activate adenylate cyclase, which is able to catalyse the conversion of large number of ATP to cAMP, hence amplifying the signal;
14. Signal transduction: cAMP, a second messenger, in turn binds and activates to large number of protein kinase A;
15. Each activated protein kinase A will initiate a sequential phosphorylation and activation of other kinases, resulting in a phosphorylation cascade;
16. At each phosphorylation step, each activated kinase is able to activate a large number of the next relay molecule / kinase;
17. At each catalytic step in the cascade, number of activated product is always greater than those in the preceding step, resulting in signal amplification;
18. Large number of glycogen phosphorylase is activated;
19. which will catalyse the hydrolysis of glycogen into glucose;
20. Cellular response: Therefore, binding of 1 glucagon to receptors will lead to the hydrolysis of large number of glycogen;

(a) Describe how the prokaryote, *E.coli*, is able to respond to varying concentrations of lactose and glucose. [8]

1. The lac operon is an inducible operon, i.e. its transcription is usually switched off;
2. The lac operon is under dual control: negative control by lac repressor and positive control by catabolite activator protein (CAP);

Negative Control

3. In the absence of lactose / when lactose concentration is low, no allolactose binds to the *lac* repressor;
 4. *lac* repressor binds to the operator site and blocks RNA polymerase from transcribing the structural genes, thus inhibits transcription;
 5. However, the binding of the repressor to operator is weak and transcription occurs at a basal level, producing small amount of lactose permease, β -galactosidase and transacetylase;
 6. In the presence of lactose, a small amount of lactose is transported into the cytoplasm via lactose permease;
 7. β -galactosidase will convert the lactose to allolactose;
 8. The cytoplasmic level of allolactose gradually rise until allolactose binds to the repressor, which has four identical subunits, each one recognising a single allolactose molecule;
 9. This results in a conformational change of the repressor, preventing it from binding to the operator site;
 10. RNA polymerase is then able to transcribe the structural genes, and the mRNA subsequently translated to form enzymes that catabolise lactose molecules;
- 7 marks max

Positive Control

11. When intracellular concentration of glucose is low, cAMP accumulates and binds to CAP;
 12. CAP is activated and binds to CAP site and enhances the rate of transcription of the structural genes;
 13. When glucose and lactose are present, CAP remains inactive and transcription of lac operon occurs at basal rate;
- 3 marks max

(b) Explain how genetic variation in bacteria could have arisen. [8]

1. Genetic diversity in bacteria comes from two sources, spontaneous mutation and genetic recombination;

Spontaneous mutation

2. Spontaneous mutation can occur from errors during DNA replication such as insertions, deletions or substitutions;
3. Increase genetic diversity quickly in species with short generation times such as bacteria;

Genetic recombination

4. Genetic recombination occurs due to the transfer of genetic material from one bacterial cell to another via transformation, transduction and conjugation;

Transformation

5. Transformation is the alteration of a bacterial cell's genotype by the uptake of naked, foreign DNA from the surrounding environment;
6. Many bacteria possess specific cell-surface proteins that recognise DNA from related species and transport it into the cell;
7. Foreign DNA is incorporated into the genome by homologous recombination / exchange of homologous DNA segments;

Conjugation

8. A sex pilus of the F^+ bacterial cell attach to the F^- bacterial cell.
9. The pilus then retracts and a temporary cytoplasmic mating bridge is formed between the two cells;
10. One strand of F plasmid's DNA is cut at the origin of transfer, and then the DNA strand travels through the pilus from the donor cell into the recipient cell.

Transduction

Description of either generalised or specialised transduction (3marks max)

11. In generalised transduction, bacteria genes are transferred randomly from one bacterial cell to another by a temperate or virulent bacteriophage.
 12. An error in the phage lytic cycle may result in a small piece of the host cell's degraded DNA to be accidentally packaged within a phage capsid in place of the phage genome.
 13. The phage can attach to another bacterium and inject the piece of bacterial DNA acquired from the first cell. The bacterial DNA is incorporated in the recipient cell's DNA by homologous recombination.
- or

14. In specialised transduction, bacteria genes adjacent to the prophage site are transferred from one bacterial cell to another by a temperate bacteriophage.
15. An error during induction of lytic cycle may result in the prophage incorrectly excised from bacterial DNA and the excised phage DNA incorporated some bacterial DNA.
16. The phage can attach to another bacterium and inject the piece of phage DNA carrying bacterial DNA acquired from the first cell. The bacterial DNA is incorporated in the recipient cell's DNA by homologous recombination.

(c) Suggest why the influenza virus displays high genetic diversity. [4]

Antigenic shift

1. antigenic shift / reassortment of RNA segments;;
2. occurs when different viral strains/subtypes infect the same cell;; REJECT different viruses
3. leads to viruses with new combinations of haemagglutinin and neuraminidase genes;;
4. Production of new haemagglutinin / neuraminidase antigens on viral envelope;;

Antigenic drift

5. Spontaneous mutation occurring in the RNA genome;
6. Resulting in change in the specific three-dimensional conformation of viral proteins / glycoproteins;